

Лабораторная работа

Номер 10

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Приобретение практических навыков по конфигурированию SMTP-сервера в части настройки аутентификации

Логи почты /var/log/maillog

```
[nsandryushin@server.nsandryushin.net ~]$ sudo -i
[sudo] password for nsandryushin:
[root@server.nsandryushin.net ~]# tail -f /var/log/maillog
Oct 31 19:49:51 server postfix/qmgr[13684]: EFFC82044481: removed
Oct 31 19:56:04 server dovecot[13834]: pop3-login: Login: user=<nsandryushin>, method=PLAIN, rip=192.168.1.1, lip=192.168.1.1, mpid=14300, secured, session=<qdB2wXlCcubAqAEB>
Oct 31 19:56:45 server dovecot[13834]: pop3(nsandryushin)<14300><qdB2wXlCcubAqAEB>: Disconnected: Logged out top=0/0, retr=1/737, del=1/2, size=1439
Oct 31 21:48:07 server dovecot[13834]: imap-login: Login: user=<nsandryushin>, method=PLAIN, rip=192.168.1.30, lip=192.168.1.1, mpid=15740, TLS, session=<Y10zUntCMp7AqAEe>
Nov  3 17:05:32 server dovecot[1457]: master: Dovecot v2.3.21 (47349e2482) starting up for imap, pop3
Nov  3 17:05:33 server postfix/postfix-script[1596]: starting the Postfix mail system
Nov  3 17:05:33 server postfix/master[1604]: daemon started -- version 3.8.5, configuration /etc/postfix
Nov  7 23:12:34 server dovecot[1460]: master: Dovecot v2.3.21 (47349e2482) starting up for imap, pop3
Nov  7 23:12:34 server postfix/postfix-script[1597]: starting the Postfix mail system
Nov  7 23:12:35 server postfix/master[1612]: daemon started -- version 3.8.5, configuration /etc/postfix
```

Рис. 1: Логи почты /var/log/maillog

```
[nsandryushin@server.nsandryushin.net ~]$ sudo -i  
[sudo] password for nsandryushin:  
[root@server.nsandryushin.net ~]# nano /etc/dovecot/dovecot.conf  
[root@server.nsandryushin.net ~]#
```

Рис. 2: /etc/dovecot/dovecot.conf

```
GNU nano 8.1 /etc/dovecot/dovecot.conf Modified
## Dovecot configuration file

# If you're in a hurry, see http://wiki2.dovecot.org/QuickConfiguration

# "doveconf -n" command gives a clean output of the changed settings. Use it
# instead of copy&pasting files when posting to the Dovecot mailing list.

# '#' character and everything after it is treated as comments. Extra spaces
# and tabs are ignored. If you want to use either of these explicitly, put the
# value inside quotes, eg.: key = "# char and trailing whitespace "

# Most (but not all) settings can be overridden by different protocols and/or
# source/destination IPs by placing the settings inside sections, for example:
# protocol imap { }, local 127.0.0.1 { }, remote 10.0.0.0/8 { }

# Default values are shown for each setting, it's not required to uncomment
# those. These are exceptions to this though: No sections (e.g. namespace {})
# or plugin settings are added by default, they're listed only as examples.
# Paths are also just examples with the real defaults being based on configure
# options. The paths listed here are for configure --prefix=/usr
# --sysconfdir=/etc --localstatedir=/var

# Protocols we want to be serving.
protocols = imap pop3 lmtp
```

Рис. 3: /etc/dovecot/dovecot.conf

```
[root@server.nsandryushin.net ~]# nano /etc/dovecot/conf.d/10-master.conf  
[root@server.nsandryushin.net ~]# █
```

Рис. 4: /etc/dovecot/conf.d/10-master.conf


```
GNU nano 8.1 /etc/dovecot/conf.d/10-master.conf Modified
#service_count = 1

# Number of processes to always keep waiting for more connections.
#process_wn_avail = 0

# If you set service_count=0, you probably need to grow this.
#vsz_limit = $default_vsz_limit
}

service pop3-login {
  inet_listener pop3 {
    #port = 110
  }
  inet_listener pop3s {
    #port = 995
    #ssl = yes
  }
}

service submission-login {
  inet_listener submission {
    #port = 587
  }
  inet_listener submissions {
    #port = 465
  }
}

service lmtp {
  unix_listener /var/spool/postfix/private/dovecot-lmtp {
    group = postfix
    user = postfix
    mode = 0600
  }
  # Create inet listener only if you can't use the above UNIX socket
  #inet_listener lmtp {
    # Avoid making LMTP visible for the entire Internet
    #address =
    #port =
  }
}
```

Рис. 5: service lmtp

```
[root@server.nsandryushin.net ~]# postconf -e 'mailbox_transport = lmtp:unix:private/dovecot-lmtp'
[root@server.nsandryushin.net ~]# nano /etc/dovecot/conf.d/10-auth.conf
```

Рис. 6: Настройка сокета

auth_username_format

```
GNU nano 8.1 /etc/dovecot/conf.d/10-auth.conf Modified
# that '#' and '/' characters are translated to '@'.
#auth_username_translation =

# Username formatting before it's looked up from databases. You can use
# the standard variables here, eg. %Lu would lowercase the username, %n would
# drop away the domain if it was given, or "%n-AT-%d" would change the '@' into
# "-AT-". This translation is done after auth_username_translation changes.
auth_username_format = %Ln
```

Рис. 7: auth_username_format

```
[root@server.nsandryushin.net ~]# systemctl restart postfix  
[root@server.nsandryushin.net ~]# systemctl restart dovecot  
[root@server.nsandryushin.net ~]#
```

Рис. 8: Перезапуск postfix и dovecot

```
[nsandryushin@client.nsandryushin.net ~]$ echo . | mail -s "LMTP test" nsandryushin@nsandryushin.net  
[nsandryushin@client.nsandryushin.net ~]$ █
```

Рис. 9: Отправка письма

```
Nov  7 23:27:47 server postfix/smtpd[9542]: connect from client.nsandryushin.net[192.168.1.30]
Nov  7 23:27:47 server postfix/smtpd[9542]: 2C0DE20444B3: client=client.nsandryushin.net[192.168.1.30]
Nov  7 23:27:47 server postfix/cleanup[9546]: 2C0DE20444B3: message-id=<20251107232746.CDA0561B7583@client.nsandryushin.net>
Nov  7 23:27:47 server postfix/smtpd[9542]: disconnect from client.nsandryushin.net[192.168.1.30] ehlo=2 starttls=1 mail=1 rcpt=1 data=1 quit=1 commands=7
Nov  7 23:27:47 server postfix/qmgr[9477]: 2C0DE20444B3: from=<nsandryushin@client.nsandryushin.net>, size=587, nrcpt=1 (queue active)
Nov  7 23:27:47 server postfix/local[9547]: 2C0DE20444B3: passing <nsandryushin@nsandryushin.net> to transport=lmtp
Nov  7 23:27:47 server dovecot[9494]: lmtp(9550): Connect from local
Nov  7 23:27:47 server dovecot[9494]: lmtp(nsandryushin)<9550><bQ7EEH0ADm10JQAA/qG3Gg>: msgid=<20251107232746.CDA0561B7583@client.nsandryushin.net>: saved mail to INBOX
Nov  7 23:27:47 server dovecot[9494]: lmtp(9550): Disconnect from local: Logged out (state=READY)
Nov  7 23:27:47 server postfix/lmtp[9548]: 2C0DE20444B3: to=<nsandryushin@nsandryushin.net>, relay=server.nsandryushin.net[private/dovecot-lmtp], delay=0.21, delays=0.04/0.02/0.08/0.08, dsn=2.0.0, status=sent (250 2.0.0 <nsandryushin@nsandryushin.net> bQ7EEH0ADm10JQAA/qG3Gg Saved)
Nov  7 23:27:47 server postfix/qmgr[9477]: 2C0DE20444B3: removed
```

Рис. 10: Логи почты

```
[nsandryushin@server.nsandryushin.net ~]$ MAIL=~/.Maildir/ mail
s-nail version v14.9.24.  Type `?' for help
/home/nsandryushin/Maildir: 2 messages 1 new
      1 Nikita Andryushin      2025-10-31 19:49    18/705    "test
►N    2 nsandryushin@client.  2025-11-07 23:27    21/901    "LMTP test
&
```

Рис. 11: Почтовый ящик mail

```
[root@server.nsandryushin.net ~]# nano /etc/dovecot/conf.d/10-master.conf
```

Рис. 12: /etc/dovecot/conf.d/10-master.conf

Структура service auth

```
GNU nano 0.1 /etc/dovecot/conf.d/10-master.conf Modified
#process_limit = 1024
}

service submission {
    # Max. number of SMTP Submission processes (connections)
    #process_limit = 1024
}

service auth {
    unix_listener /var/spool/postfix/private/auth {
        group = postfix
        user = postfix
        mode = 0660
    }
    # auth_socket_path points to this userdb socket by default. It's typically
    # used by dovecot-lda, doveadm, possibly imap process, etc. Users that have
    # full permissions to this socket are able to get a list of all usernames and
    # get the results of everyone's userdb lookups.
    #
    # The default 0666 mode allows anyone to connect to the socket, but the
    # userdb lookups will succeed only if the userdb returns an 'uid' field that
    # matches the caller process's UID. Also if caller's uid or gid matches the
    # socket's uid or gid the lookup succeeds. Anything else causes a failure.
    #
    # To give the caller full permissions to lookup all users, set the mode to
    # something else than 0666 and Dovecot lets the kernel enforce the
    # permissions (e.g. 0777 allows everyone full permissions).
    unix_listener auth-userdb {
        mode = 0600
        user = dovecot
        #group =
    }

    # Postfix smtp-auth
    #unix_listener /var/spool/postfix/private/auth {
    #    mode = 0666
    #}

    # Auth process is run as this user.
    #user = $default_internal_user
}
```

Рис. 13: Структура service auth

Настройка почты в postfix

```
[root@server.nsandryushin.net ~]# postconf -e 'smtpd_sasl_type = dovecot'
[root@server.nsandryushin.net ~]# postconf -e 'smtpd_sasl_path = private/auth'
[root@server.nsandryushin.net ~]# postconf -e 'smtpd_recipient_restrictions = reject_unknown_recipient_domain, permit_mynetworks, reject_non_fqdn_recipient, reject_unauth_destination, reject_unverified_recipient, permit'
[root@server.nsandryushin.net ~]# postconf -e 'mynetworks = 127.0.0.0/8'
[root@server.nsandryushin.net ~]#
```

Рис. 14: Настройка почты в postfix

```
[root@server.nsandryushin.net ~]# nano /etc/postfix/master.cf
```

Рис. 15: /etc/postfix/master.cf

Включение авторизации

```
GNU nano 2.9.1 /etc/postfix/master.cf
#
# Postfix master process configuration file. For details on the format
# of the file, see the master(5) manual page (command: "man 5 master" or
# on-line: http://www.postfix.org/master.5.html).
#
# Do not forget to execute "postfix reload" after editing this file.
#
# =====
# service type private unpriv chroot wakeup maxproc command + args
# (yes) (yes) (no) (never) (100)
# =====
smtp inet n - n - - smtpd
-o smtpd_sasl_auth_enable=yes
-o smtpd_recipient_restrictions=reject_non_fqdn_recipient, reject_unknown_recipient_domain, permit_sasl_authenticated
smtp inet n - n - - smtpd
#smtpd pass - - n - - smtpd
#dnsmlog unix - - n - - dnsmlog
#tssproxy unix - - n - - tssproxy
# Choose one: enable submission for loopback clients only, or for any client.
#127.0.0.1:submission inet n - n - - smtpd
#submission inet n - n - - smtpd
# -o syslog_name=postfix/submission
# -o smtpd_tls_security_level=encrypt
# -o smtpd_sasl_auth_enable=yes
# -o smtpd_tls_auth_only=yes
# -o local_header_rewrite_clients=static:all
# -o smtpd_reject_unlisted_recipient=no
# Instead of specifying complex smtpd_<xxx>_restrictions here,
# specify "smtpd_<xxx>_restrictions=$muu_<xxx>_restrictions"
# here, and specify muu_<xxx>_restrictions in main.cf (where
# "<xxx>" is "client", "helo", "sender", "relay", or "recipient").
# -o smtpd_client_restrictions=
# -o smtpd_helo_restrictions=
# -o smtpd_sender_restrictions=
# -o smtpd_relay_restrictions=
# -o smtpd_recipient_restrictions=permit_sasl_authenticated, reject
# -o milter_macro_daemon_name=ORIGINATING
# Choose one: enable submissions for loopback clients only, or for any client.
#127.0.0.1:submissions inet n - n - - smtpd
#submissions inet n - n - - smtpd
# -o syslog_name=postfix/submissions
# -o smtpd_tls_wrappermode=yes
# -o smtpd_sasl_auth_enable=yes
```

Рис. 16: Включение авторизации

```
[root@server.nsandryushin.net ~]# systemctl restart postfix  
[root@server.nsandryushin.net ~]# systemctl restart dovecot  
[root@server.nsandryushin.net ~]#
```

Рис. 17: Перезапуск postfix и dovecot

Установка Telnet

```
[nsandryushin@client nsandryushin.net ~]$ sudo -i
[sudo] password for nsandryushin:
[root@client nsandryushin.net ~]# dnf -y install telnet
Extra Packages for Enterprise Linux 10 - x86_64                7.8 kB/s | 40 kB    00:05
Extra Packages for Enterprise Linux 10 - x86_64                2.1 MB/s | 4.8 MB    00:02
Rocky Linux 10 - BaseOS                                       815 B/s | 4.3 kB    00:05
Rocky Linux 10 - BaseOS                                       2.1 MB/s | 23 MB    00:11
Rocky Linux 10 - AppStream                                    7.4 kB/s | 4.3 kB    00:00
Rocky Linux 10 - AppStream                                    1.2 MB/s | 2.2 MB    00:01
Rocky Linux 10 - CRB                                          14 kB/s | 4.3 kB    00:00
Rocky Linux 10 - CRB                                          698 kB/s | 535 kB    00:00
Rocky Linux 10 - Extras                                       9.7 kB/s | 3.1 kB    00:00
Rocky Linux 10 - Extras                                       12 kB/s | 5.5 kB    00:00
Dependencies resolved.
=====
Package             Architecture      Version           Repository         Size
-----
Installing:
telnet              x86_64            1:0.17-94.el10    appstream          62 k
Transaction Summary
=====
Install 1 Package

Total download size: 62 k
Installed size: 109 k
Downloading Packages:
telnet-0.17-94.el10.x86_64.rpm                                497 kB/s | 62 kB    00:00
-----
Total                                                         147 kB/s | 62 kB    00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Installing    : telnet-1:0.17-94.el10.x86_64    1/1
  Running scriptlet: telnet-1:0.17-94.el10.x86_64  1/1

Installed:
telnet-1:0.17-94.el10.x86_64                                I

Complete!
[root@client nsandryushin.net ~]# █
```

Рис. 18: Установка Telnet

Авторизация по telnet

```
[nsandryushin@client.nsandryushin.net ~]$ printf 'nsandryushin\x00nsandryushin\x00123456' | base64
bnNhbmRyeXVzaGluAG5zYW5kcml1c2hpbGxMjM0NTY=
[nsandryushin@client.nsandryushin.net ~]$ telnet server.nsandryushin.net 25
Trying 192.168.1.1...
Connected to server.nsandryushin.net.
Escape character is '^J'.
220 server.nsandryushin.net ESMTP Postfix
EHLO test
250-server.nsandryushin.net
250-PIPELINING
250-SIZE 10240000
250-VRFY
250-ETRN
250-STARTTLS
250-AUTH PLAIN
250-ENHANCEDSTATUSCODES
250-8BITIME
250-DSN
250-SMTPUTF8
250 CHUNKING
AUTH PLAIN bnNhbmRyeXVzaGluAG5zYW5kcml1c2hpbGxMjM0NTY=
235 2.7.0 Authentication successful
exit
500 5.5.2 Error: command not recognized
^C^C
quit
Connection closed by foreign host.
[nsandryushin@client.nsandryushin.net ~]$ █
```

Рис. 19: Авторизация по telnet

Настройка postfix

```
[root@server.nsandryushin.net ~]# cp /etc/pki/dovecot/certs/dovecot.pem /etc/pki/tls/certs
[root@server.nsandryushin.net ~]# cp /etc/pki/dovecot/private/dovecot.pem /etc/pki/tls/private
[root@server.nsandryushin.net ~]# postconf -e 'smtpd_tls_cert_file=/etc/pki/tls/certs/dovecot.pem'
[root@server.nsandryushin.net ~]# postconf -e 'smtpd_tls_key_file=/etc/pki/tls/private/dovecot.pem'
[root@server.nsandryushin.net ~]# postconf -e 'smtpd_tls_session_cache_database = btree:/var/lib/postfix/smtpd_scache'
[root@server.nsandryushin.net ~]# postconf -e 'smtpd_tls_security_level = may'
[root@server.nsandryushin.net ~]# postconf -e 'smtp_tls_security_level = may'
[root@server.nsandryushin.net ~]# nano /etc/postfix/master.cf
```

Рис. 20: Настройка postfix


```
GNU nano 8.1 /etc/postfix/master.cf Modified
#
# Postfix master process configuration file. For details on the format
# of the file, see the master(5) manual page (command: "man 5 master" or
# on-line: http://www.postfix.org/master.5.html).
#
# Do not forget to execute "postfix reload" after editing this file.
#
# =====
# service type private unpriv chroot wakeup maxproc command + args
#          (yes)   (yes)   (no)   (never) (100)
# =====
smtp      inet  n       -       n       -       -       smtpd
#smtp     inet  n       -       n       -       1       postscreen
#smtpd    pass  -       -       n       -       -       smtpd
#dnsblog  unix  -       -       n       -       0       dnsblog
#tlsproxy unix  -       -       n       -       0       tlsproxy
# Choose one: enable submission for loopback clients only, or for any client.
#127.0.0.1:submission inet n - n - - smtpd
submission inet n       -       n       -       -       smtpd
  -o smtpd_tls_security_level=encrypt
  -o smtpd_sasl_auth_enable=yes
  -o smtpd_recipient_restrictions=reject_non_fqdn_recipient,reject_unknown_recipient_domain,permit_sasl_>
```

Рис. 21: /etc/postfix/master.cf

Настройка firewall

```
[root@server.nsandryushin.net ~]# firewall-cmd --get-services
0-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr amanda-client amanda-k5-client amqp amqps anno-1602 a
nno-1800 apcupsd aseqnet audit ausweisapp2 bacula bacula-client bareos-director bareos-filedemon bareos-
storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter
ceph-mon cfengine checkmk-agent civilization-iv civilization-v cockpit collectd condor-collector cratedb
ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-quic dns-over-tls docker
-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server factorio finger foreman fore
man-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-clien
t ganglia-master git gpsd grafana gre high-availability http http3 https ident imap inaps iperf2 iperf3 i
pps ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadm5 kdeconnect kerberos kibana klogon kpas
swd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-man
ager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-work
er kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llennr llennr-c
lient llennr-tcp llennr-udp managiesieve matrix ndns memcache minecraft minidlna mndp mongodb mosh mountd sp
d mqtt mqtt-tls ns-wbt mssql murmur mysql nbd nebula need-for-speed-most-wanted netbios-ns netdata-dashbo
ard nfs nfs3 nmap-0183 nripe ntp nut opentelemetry openvpn ovirt-inagelo ovirt-storageconsole ovirt-vmcons
ole plex pmed papproxy pwebapi pwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporte
r proxy-dhcp ps2link ps3netstrv ptp pulseaudio puppetmaster quassel radius radsec rdp redis redis-sentinel
rootd rpc-bind rquotad rsh rsysd rtsp salt-master samba samba-client samba-dc sane settlers-history-col
lection sip sipps slimevr slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap spiderOak-lans
ync spotify-sync squid ssdp ssh ssh-custom statshv steam-lan-transfer steam-streaming stellaris stronghol
d-crusader stun stuns submission supertuxkart svdrp svn syncthing syncthing-gui syncthing-relay synergy s
yscomlan syslog syslog-tls telnet tentacle terraria tftp tile38 tinc tor-socks transmission-client turn t
urns upnp-client vdsu vnc-server vrrp warpinator wben-http wben-https wireguard ws-discovery ws-discovery
-client ws-discovery-host ws-discovery-tcp ws-discovery-udp wssd wssd-http wsmn wsmns xdmcp xmp-bosh x
mp-client xmp-local xmp-server zabbix-agent zabbix-java-gateway zabbix-server zabbix-trapper zabbix-we
b-service zero-k zerotier
[root@server.nsandryushin.net ~]# firewall-cmd --add-service=smtp-submission
success
[root@server.nsandryushin.net ~]# firewall-cmd --add-service=smtp-submission --permanent
success
[root@server.nsandryushin.net ~]# firewall-cmd --reload
success
[root@server.nsandryushin.net ~]# systemctl restart postfix
[root@server.nsandryushin.net ~]#
```

Рис. 22: Настройка firewall

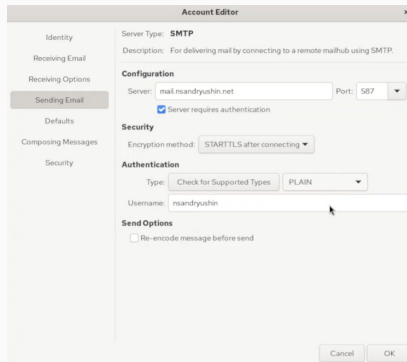
```
[root@server.nsandryushin.net ~]# openssl s_client -starttls smtp -crlf -connect server.nsandryushin.net:587
Connecting to 192.168.1.1
CONNECTED(SSLv3)
depth=0 OU=IMAP server, CN=imap.example.com, emailAddress=postmaster@example.com
verify error:num=10:self-signed certificate
verify return:1
depth=0 OU=IMAP server, CN=imap.example.com, emailAddress=postmaster@example.com
verify return:1
---
Certificate chain
 0 s:OU=IMAP server, CN=imap.example.com, emailAddress=postmaster@example.com
 1:OU=IMAP server, CN=imap.example.com, emailAddress=postmaster@example.com
 a:PKEY: rsaEncryption, 3072 (bit); sigalg: RSA-SHA256
 v:NotBefore: Oct 31 19:29:08 2025 GMT, NotAfter: Oct 31 19:29:08 2026 GMT
---
Server certificate
-----BEGIN CERTIFICATE-----
MIIECjCCAtgqAwIBAgIU0Dn1bNnGdURybFlwufqKwDOYJKoZi3hvcNADEL
BQAwMEUmbG91UECwLSU1BUCBzKJZlZXIvOTAXBgNVBAMEM01TYXN0XkhbXBs
ZSS5b28xJTAjBgkqhkiG9w0BBQcGFncBvc3RlYXN0ZXJAZXhhbBzZSS5b28xhcn
NjUuMDEwMTkxOTAA8bChMjYkMDRkMTkyOTA4BjBVMRQwQWY0VGVQLDA1JTUwQ1h0
cnZlcjZBRBGAUUEwQW81bG51eGFCc0x1LWwvbnRlMCMQCSqGSIIS3DQEJARYW
c09udGhlc3Rlc0k1bG91dG91LWwvbnRlMCMQCSqGSIIS3DQEJAKoZi3hvcNADEL
AYCggGBAMCkXUjMjG0M0ASIAQ/paDQ+QnbBHSaortR8p9D0M0NS5epta1YoyLzta
w8jJrpFAnyP1+0y3yecaBjHd4890H6nWH06IVhy1chxSunGtJoa6ZK+y2vn1DL6
LMOz1J1/LP0kFvRqD1ngDmxCp0M9gcK4cECs+LXFqBfZgg8Kf0L4K1UHQjQje
5wvEB0dXUCCIEpKRC2Kcz2rBh3H6KjW6KqL2qT777p7PJaqlqCULvutHHSFwV
jy91tp15Ssejb795sP4UQ/1MwKWSRjdeva1gcZ52A819K1MF8KVLH8ScFupnd
/NL2wYJi2z5D0n1V1/07H14ECq8F7JH0T0v7qzs4EeMhAw12V1kxpl+tgHCKZD
qso39feyqf3r/KC7xquLb19TxXhFDMi07P15TPyq2yZVReZyOpR/1zFPecIBY
9kR8uYcxa729VPV5PAXNHzKvG8yH0jJdyq/0ydzrMDNgat081Umcg4Q3hBYV+
0ev8FAQbAwIQAQABu0zAmJjAR8gJghkqBh+nCAGE5BAPC8kAw0YDUR088YETmd
PYmepPK1t3BUDYQr0pVM0OC5qS5Ib3D0EB0UAA41BgQATkxAU88xQLdH
cW0gIE4UTrcuGefahV185fnPacKwCX8nztwae1ZVC+9fslnmr9raH3v76eIZVvy
14wXvt0zV0J4uq0vN1SA9HY72kuXeh08qZ/dsx25z5bNP5u2zBBRkXervjU1XU
wucvX1qk09MGS+y2z8y8j0bJ6J2Tf01L13qXlnL5861GS+nB83rANZ5+AtvD
L168u4t1L8T4p10LURHj03y03agj/5D44P8U0dGSRV98tcTgF8jJfF030d7C4F
n5Ldsd180JdrtT3WAdZn8q07p+n4MQSg3pKX+nvB01snT803eP88je8Wc4M34Q
Zfz9T9E/81u+K5uBz4h61P7ChnZCZDUzrakGC/YkLhJp1K8h+q4d1h5dFxt0d
sK1CKPdpb0wA90VENze4YK/QVhNKTQRUL7hZ19hp8J98e6n18VVKwELT1M8/U
eRgAK+1Kd1K9w416g2hwZEqot4Kuqx03fCnSU0g5PbS1Z1hAi=
-----END CERTIFICATE-----
```

Рис. 23: openssl

```
EHLO test
250-server.nsandryushin.net
250-PIPELINING
250-SIZE 10240000
250-VERFY
250-ETRN
250-AUTH PLAIN
250-ENHANCEDSTATUSCODES
250-8BITMIME
250-DSN
250-SMTPUTF8
250 CHUNKING
AUTH PLAIN bnNhbMRYeXVzaGluAG5zYW5kcml1c2hpbGxMjM0NTY=
235 2.7.0 Authentication successful
```

Рис. 24: Авторизация

Настройка smtp в клиенте



The screenshot shows the 'Account Editor' window with the 'Sending Email' tab selected. The configuration is for an SMTP server. The 'Server' field is 'mail.nsandyushin.net' and the 'Port' is '587'. The 'Server requires authentication' checkbox is checked. The 'Security' section shows 'Encryption method' set to 'STARTTLS after connecting'. The 'Authentication' section shows 'Type' set to 'PLAIN' and 'Username' set to 'nsandyushin'. The 'Send Options' section has 'Re-encode message before send' unchecked. The 'Cancel' and 'OK' buttons are at the bottom right.

Account Editor

Identity

Receiving Email

Receiving Options

Sending Email

Defaults

Composing Messages

Security

Server Type: **SMTP**

Description: For delivering mail by connecting to a remote mailhub using SMTP.

Configuration

Server: mail.nsandyushin.net Port: 587

☒ Server requires authentication

Security

Encryption method: STARTTLS after connecting

Authentication

Type: Check for Supported Types PLAIN

Username: nsandyushin

Send Options

☐ Re-encode message before send

Cancel OK

Рис. 25: Настройка smtp в клиенте

Отправка письма

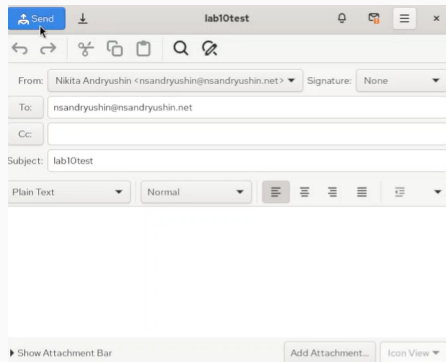


Рис. 26: Отправка письма

Письмо в ящике

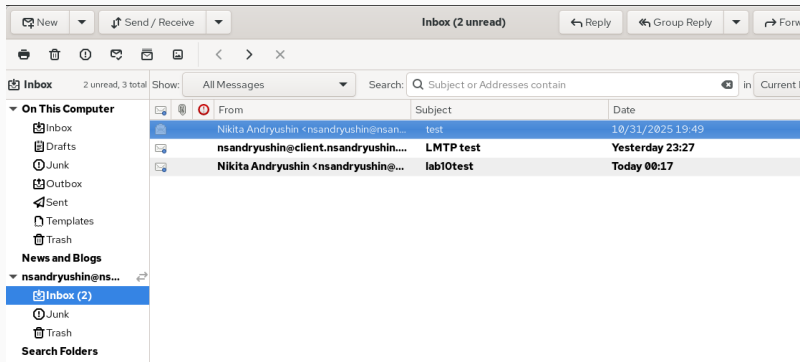


Рис. 27: Письмо в ящике

```
Nov  8 00:17:14 server postfix/smtpd[11114]: B4E8A2044481: client=client.nsandryushin.net[192.168.1.30],  
sasl_method=PLAIN, sasl_username=nsandryushin  
Nov  8 00:17:14 server postfix/cleanup[11145]: B4E8A2044481: message-id=<32e371f2b28536190d25e6749a146638  
61678ffe.camel@nsandryushin.net>  
Nov  8 00:17:14 server postfix/qmgr[11103]: B4E8A2044481: from=<nsandryushin@nsandryushin.net>, size=587,  
nrcpt=1 (queue active)  
Nov  8 00:17:14 server postfix/smtpd[11114]: disconnect from client.nsandryushin.net[192.168.1.30] ehlo=2  
starttls=1 auth=1 mail=1 rcpt=1 data=1 rset=1 quit=1 commands=9  
Nov  8 00:17:14 server postfix/local[11146]: B4E8A2044481: passing <nsandryushin@nsandryushin.net> to tra  
nsport=lmtp  
Nov  8 00:17:14 server dovecot[10024]: lmtp(11148): Connect from local  
Nov  8 00:17:14 server dovecot[10024]: lmtp(nsandryushin)<11148><C9D1MAqMDmmMKwAA/qG3Gg>: msgid=<32e371f2  
b28536190d25e6749a14663861678ffe.camel@nsandryushin.net>: saved mail to INBOX  
Nov  8 00:17:14 server dovecot[10024]: lmtp(11148): Disconnect from local: Logged out (state=READY)  
Nov  8 00:17:14 server postfix/lmtp[11147]: B4E8A2044481: to=<nsandryushin@nsandryushin.net>, relay=serve  
r.nsandryushin.net[private/dovecot-lmtp], delay=0.12, delays=0.04/0.02/0.05/0.01, dsn=2.0.0, status=sent  
(250 2.0.0 <nsandryushin@nsandryushin.net> C9D1MAqMDmmMKwAA/qG3Gg Saved)  
Nov  8 00:17:14 server postfix/qmgr[11103]: B4E8A2044481: removed
```

Рис. 28: Логи почты


```
[root@server.nsandryushin.net ~]# cd /vagrant/provision/server
[root@server.nsandryushin.net server]# cp -R /etc/dovecot/dovecot.conf
cp: missing destination file operand after '/etc/dovecot/dovecot.conf'
Try 'cp --help' for more information.
[root@server.nsandryushin.net server]# cp -R /etc/dovecot/dovecot.conf /vagrant/provision/server/mail/etc
/dovecot/
cp: overwrite '/vagrant/provision/server/mail/etc/dovecot/dovecot.conf'? y
[root@server.nsandryushin.net server]# cp -R /etc/dovecot/conf.d/10-master.conf /vagrant/provision/server
/mail/etc/dovecot/conf.d/
[root@server.nsandryushin.net server]# cp -R /etc/dovecot/conf.d/10-auth.conf /vagrant/provision/server/m
ail/etc/dovecot/conf.d/
cp: overwrite '/vagrant/provision/server/mail/etc/dovecot/conf.d/10-auth.conf'? y
[root@server.nsandryushin.net server]# mkdir -p /vagrant/provision/server/mail/etc/postfix/
[root@server.nsandryushin.net server]# cp -R /etc/postfix/master.cf /vagrant/provision/server/mail/etc/po
stfix/
[root@server.nsandryushin.net server]# nano /vagrant/provision/server/mail.sh
```

Рис. 29: vagrant

mail.sh для сервера

```
❯ cat /usr/share/doc/mailutils-2.12.2/mail.sh
#!/bin/bash
echo "Provisioning script 50"
echo "Install needed packages"
dnf -y install postfix
dnf -y install dovecot
dnf -y install telnet
dnf -y install s-nail
echo "Copy configuration files"
cp -R /usr/share/doc/postfix-3.10.0/* /etc
chown -R root:root /etc/postfix
restoredcon -vR /etc
echo "Configure firewall"
firewall-cmd --add-service smtp --permanent
firewall-cmd --add-service pop3 --permanent
firewall-cmd --add-service pop3s --permanent
firewall-cmd --add-service imap --permanent
firewall-cmd --add-service imaps --permanent
firewall-cmd --add-service smtp-submission --permanent
firewall-cmd --reload
echo "Start postfix service"
systemctl enable postfix
systemctl start postfix
echo "Configure postfix"
postconf -e 'mydomain = nsandryushin.net'
postconf -e 'myorigin = $mydomain'
postconf -e 'inet_protocols = ipv4'
postconf -e 'inet_interfaces = all'
postconf -e 'mydestination = $myhostname, localhost.$mydomain, localhost.$mydomain'
postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
echo "Configure postfix for dovecot"
postconf -e 'home_mailbox = Maildir/'
echo "Configure postfix for auth"
postconf -e 'smtpd_sasl_type = dovecot'
postconf -e 'smtpd_sasl_path = private/auth'
postconf -e 'smtpd_recipient_restrictions = reject_unknown_recipient_domain, permit_mynetworks, reject_n...'
postconf -e 'mynetworks = 127.0.0.0/8'
echo "Configure postfix for SMTP over TLS"
cp /etc/pki/dovecot/certs/dovecot.pem /etc/pki/tls/certs
cp /etc/pki/dovecot/private/dovecot.pem /etc/pki/tls/private
postconf -e 'smtpd_tls_cert_file=/etc/pki/tls/certs/dovecot.pem'
postconf -e 'smtpd_tls_key_file=/etc/pki/tls/private/dovecot.pem'
postconf -e 'smtpd_tls_session_cache_database = btree:/var/lib/postfix/smtpd_scache'
postconf -e 'smtpd_tls_security_level = may'
postconf -e 'smtp_tls_security_level = may'
```

Рис. 30: mail.sh для сервера

mail.sh для клиента

```
GNU nano 8.1                               /vagrant/provision/client/mail.sh    Modified
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
echo "Configure postfix"
postconf -e 'inet_protocols = ipv4'
echo "Start postfix service"
systemctl enable postfix
systemctl start postfix

dnf -y install evolution
dnf -y install telnet
```

Рис. 31: mail.sh для клиента

В результате выполнения лабораторной работы были получены навыки продвинутой настройки smtp и авторизации